

Project TARS - Project TODO and Verification

PRIMARY PROJECT CONTROL DOCUMENT

Status: Version 0.2 - Primary implementation, test, and verification checklist

Date: 2026-08-20

Owner: Project TARS implementation and verification workstreams

Companion documents: VAD architecture, FreeRTOS execution plan, firmware build guide, P4-to-Pi audio protocol, and VED Training start page

1. Purpose

This is the master executable checklist for implementing and verifying Project TARS across the P4 firmware, physical hardware, Raspberry Pi 5 services, facial gestures, speech pipeline, integration tests, and release gates. The current critical path begins with PC recordings and physical P4 microphone validation, then proceeds through VAD, Pi services, and full-system testing.

This file is the primary day-to-day project document for deciding what to do next, running tests, recording evidence, and deciding whether a goal is truly complete. Companion documents explain design details; completion status belongs here.

It deliberately separates three functions:

Function	Owner	Work performed
Voice activity detection (VAD)	P4	Detect speech start/end; calibrate thresholds and endpoint timing
Speech-to-text (STT)	Pi 5	Convert bounded speech audio into words
Speaker verification	Pi 5	Enroll and compare the operator's voice using a pretrained embedding model

"VAD training" in informal project discussions means corpus-based calibration and testing. VAD is not trained to recognize the operator. Speaker recognition uses enrollment and held-out testing, not a newly trained neural model.

2. How to Use This Checklist

For every development or test session:

1. Start with the first incomplete item on the critical path unless a blocker is recorded beside it.
2. Run the relevant build or validation command before hardware testing.
3. Preserve the evidence named under that goal: logs, reports, WAV metadata, counters, screenshots, or protocol traces.
4. Tick an item only after its stated behavior has been observed. A successful

compile proves the code builds; it does not prove the microphone, LCD, Wi-Fi, PSRAM, or timing on physical hardware.

5. Record measured values and the test date in this document or a linked report.
6. Update the goal summary whenever a goal changes state.

All actionable checkboxes from the firmware roadmap, hardware inventory, and speech-runtime evaluation are consolidated here. Those documents remain the source for architecture, rationale, specifications, and measured facts, but they are not separate progress trackers.

Standard clean firmware verification from the repository root:

```
.\tools\build-firmware.ps1 -Clean
Test-Path .\build\project_tars.bin
Test-Path .\build\bootloader\bootloader.bin
```

Both Test-Path commands must return True, and the build must end with `Project build complete`. See the firmware build guide for the sandbox/toolchain explanation and configurable installation paths.

Current next test

The next critical-path activity is G1 physical P4 microphone validation:

TODO	Connect the P4 board and record its serial port and board revision.
TODO	Run the clean firmware verification command above.
TODO	Flash the generated firmware and retain the complete boot/serial log.
TODO	Confirm the endpoint self-test reports passed during startup.
TODO	Confirm ES8311 initialization succeeds and <code>audio_rx_task</code> remains alive.
TODO	Speak normally, quietly, and loudly while recording peak, RMS, clipping, queue depth, dropped-frame, sequence-gap, read-error, and stack-watermark values.
TODO	Add the measurements under G1 before marking hardware items complete.

3. Current Baseline

Prepared

DONE	VAD architecture and endpointing plan documented.
DONE	FreeRTOS task/core allocation documented.
DONE	Private PC recording workspace created under VED Training.
DONE	Multi-tone recording plan and manifest templates created.
DONE	WAV format/clipping checker created.
DONE	Corpus, VAD-boundary, and speaker-score evaluation tools created.
DONE	Raw recordings excluded from Git.
DONE	Initial 16 kHz P4 microphone diagnostic source implemented.
DONE	Clean ESP-IDF 6.0.2 firmware build verified on 2026-08-20.
DONE	Bounded 500 ms producer/consumer audio queue implemented.

DONE	Backend-independent endpoint state machine and on-device transition self-test implemented.
DONE	Version 1 P4-to-Pi audio/control protocol documented.

Verified development evidence

Date	Verification	Result
2026-08-20	build-firmware.ps1 -Clean	Passed: application 2131/2131 and bootloader 174/174
2026-08-20	Incremental build-firmware.ps1	Passed: firmware image regenerated successfully
2026-08-20	Endpoint synthetic startup test	Compiled into firmware; physical startup log still required

Not yet proven

TODO	Real PC corpus has been recorded and validated.
TODO	P4 onboard microphone capture has been saved as a valid WAV.
TODO	ESP-SR has been integrated into the firmware.
TODO	Automatic P4 speech start/end events have been measured.
TODO	P4-to-Pi audio streaming has been proven end to end.
TODO	Pi 5 STT and speaker enrollment have been benchmarked.

Checked items above mean the planning/tooling exists; they do not claim that the corresponding runtime feature works on hardware.

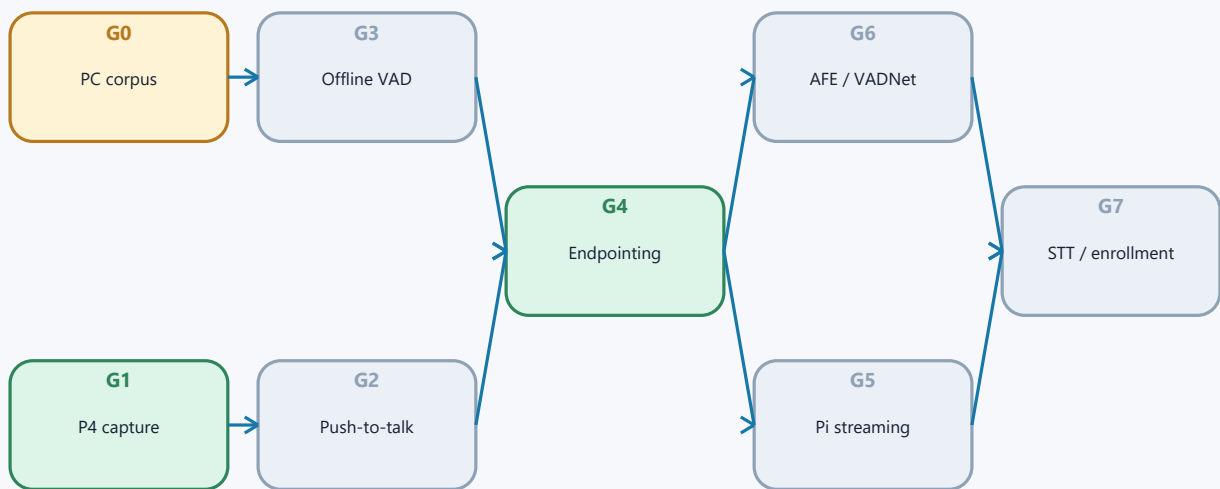
4. Goal Summary and Critical Path

Goal	Outcome	Depends on	Status
G0	PC recording workflow proven	None	Ready to start
G1	P4 microphone capture proven	Board audio driver	In progress - hardware test next
G2	Push-to-talk path proven	G1	Not started
G3	Standalone VAD works offline	G0, ESP-SR dependency	Not started
G4	P4 endpointing preserves full utterances	G1, G3	In progress - state machine implemented
G5	VAD-bounded audio reaches Pi 5	G2, G4, Wi-Fi session	Not started
G6	AFE/VADNet production path evaluated	G4	Not started
G7	STT and speaker enrollment proven	G0, Pi 5 runtime	Not started
G8	Face states follow the voice pipeline	G4, gesture renderer	Not started
G9	Concurrent system passes soak test	G5-G8	Not started

Critical path:

Critical path to full-system proof

Green: active | Amber: ready | Grey: queued



5. G0 - Prove the PC Recording Workflow

Goal: Produce correctly formatted, private recordings that exercise natural changes in the operator's voice.

TODO	Install Audacity from the official source.
TODO	Run VED Training/tools/setup.ps1.
TODO	Select the intended PC microphone and record in mono at 16 kHz.
TODO	Record room silence, normal speech, and quiet speech test clips.
TODO	Run VED Training/tools/check_wav.py against those clips.
TODO	Correct sample rate, channel count, encoding, or gain problems.
TODO	Complete session 001 from VED Training/plans/recording-plan.csv.
TODO	Complete session 002 on another day.
TODO	Complete session 003 for prompts requiring a third repetition.
TODO	Populate the private corpus manifest with actual transcripts.
TODO	Run validate_corpus.py with zero format or metadata errors.
TODO	Back up the private corpus securely without adding it to Git.

Evidence	Private validation report plus session notes.
Exit condition	Every accepted clip is mono, 16-bit, 16 kHz PCM WAV, unclipped, correctly named, and assigned to only one data split.

6. G1 - Prove P4 Onboard Microphone Capture

Goal: Capture the actual signal that the deployed system will hear.

TODO	Confirm ES8311 I2C communication and codec identity/configuration.
DONE	Configure MCLK, BCLK, LRCK, and I2S receive routing in the diagnostic.
DONE	Start with 16 kHz, signed 16-bit, mono application audio.
DONE	Drain fixed 20 ms reads from a dedicated <code>audio_rx_task</code> on Core 1.
DONE	Add read-error, peak, RMS, clipping, and stack-watermark counters.
TODO	Confirm the diagnostic configuration and readings on physical hardware.
DONE	Add a bounded producer/consumer ring before networking or file output.
TODO	Save a bounded diagnostic WAV without blocking the DMA task.
TODO	Record P4 silence, normal speech, quiet speech, and loud speech.
TODO	Validate the P4 WAV files with the same PC checker.
TODO	Document actual gain, noise floor, DC offset, clipping, and channel map.
TODO	Run a one-hour capture soak test.
Evidence	Valid P4 diagnostic WAVs and health-counter log.
Exit condition	No unexplained DMA overruns, watchdog resets, malformed WAVs, or clipped normal speech during the soak test.

7. G2 - Prove Push-to-Talk Before Automatic VAD

Goal: Isolate microphone, buffering, networking, and STT from VAD tuning.

TODO	Add an explicit start/stop input or temporary software command.
TODO	Allocate bounded PCM blocks and a PSRAM history ring.
TODO	Emit <code>AUDIO_START</code> , ordered <code>AUDIO_CHUNK</code> , and <code>AUDIO_END</code> messages.
TODO	Receive and save the stream on a PC/Pi test receiver.
TODO	Compare received PCM length and checksum/counters with the source.
TODO	Run Pi 5 STT against at least 20 push-to-talk utterances.
TODO	Record capture-to-first-transcript and end-to-final-transcript latency.
TODO	Verify clean cancellation, timeout, and reconnect behavior.
Evidence	Receiver logs, saved WAVs, STT results, and latency report.

Exit condition	Repeated utterances arrive without missing/reordered audio and produce stable transcripts before automatic detection is introduced.
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8. G3 - Integrate and Test Standalone ESP-SR VAD Offline

Goal: Establish a minimal P4-compatible VAD baseline using repeatable audio.

TODO	Select and pin an exact compatible espressif/esp-sr version.
TODO	Record component/model licensing and redistribution obligations.
TODO	Compile a host or P4 recorded-file VAD harness.
TODO	Feed 20 ms frames: 320 samples, 640 bytes, 16 kHz mono PCM.
TODO	Export detected state and predicted speech boundaries to CSV.
TODO	Test normal, quiet, loud, fast, slow, tired, and energetic speech.
TODO	Test silence, fan, keyboard, and representative room noise.
TODO	Run <code>VED Training/tools/evaluate_vad.py</code> .
TODO	Record results separately by tone, distance, and noise condition.
TODO	Select an initial VAD mode/threshold without using the final test split.

Evidence	Reproducible build/version record, prediction CSV, and evaluation report.
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Exit condition	Validation targets in the VAD plan are met or the failure is documented with a specific mitigation experiment.
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9. G4 - Implement P4 Endpointing and Pre-Roll

Goal: Turn frame-level speech/silence states into complete utterances.

TODO	Implement a backend-independent <code>tars_vad</code> interface.
DONE	Implement the <code>tars_endpoint</code> state machine separately from the backend.
TODO	Add at least 500 ms of bounded pre-roll.
DONE	Add start confirmation and end-of-speech hangover/hysteresis.
DONE	Add maximum utterance and no-speech timeouts.
TODO	Preserve pre-trigger frames through the VAD cache where supported.
DONE	Add deterministic synthetic VAD sequences to the on-device startup self-test.
TODO	Verify short acknowledgements such as "yes", "no", and "pause".
TODO	Verify that the first phoneme is not clipped.
TODO	Recover cleanly after ring overrun or backend failure.

TODO	Retain push-to-talk as the fallback mode.
Evidence	Unit-test output, audio boundary overlays, and failure-injection results.
Exit condition	At least 98% normal-desk detection, 95% across planned tone conditions, and no more than 150 ms leading clipping on over 99% of accepted validation clips.

10. G5 - Stream VAD-Bounded Audio to the Pi 5

Goal: Make P4 endpoint events the primary turn boundaries for conversation.

TODO	Connect endpoint output to the existing bounded uplink path.
TODO	Establish the trusted Wi-Fi LAN service path between the P4 and Pi 5.
TODO	Send pre-roll before live chunks without duplication.
TODO	Include stream ID, sequence, format, timestamps, and end reason.
TODO	Apply explicit backpressure and maximum-buffer policies.
TODO	Preserve local responsiveness during Wi-Fi stalls.
TODO	Verify automatic reconnect without replaying stale utterances.
TODO	Allow Pi-side endpointing only as a secondary diagnostic/safety check.
TODO	Test 100 consecutive automatic utterances.
TODO	Measure P4-to-Pi latency and throughput under idle and inference load.
TODO	Record the first complete capture-to-transcript-to-response latency.
Evidence	Protocol trace, sequence-gap counter, latency distribution, and saved receiver audio.
Exit condition	No unreported audio gaps, stale replay, or UI stalls during normal operation and controlled reconnect tests.

11. G6 - Evaluate AFE/VADNet and Echo Handling

Goal: Decide whether ESP-SR AFE/VADNet is the production audio front end.

TODO	Integrate AFE feed/fetch tasks without violating frame contracts.
TODO	Compare AFE/VADNet against standalone VAD using the same corpus.
TODO	Measure Core 1 load, internal RAM, PSRAM, and end-to-end latency.
TODO	Verify VAD cache/pre-roll handling.
TODO	Add noise suppression only as a measured experiment.
TODO	Begin half-duplex with capture muted or gated during TARS playback.
TODO	Route playback reference audio before enabling AEC.

TODO	Evaluate AEC using real speaker-to-microphone coupling.
TODO	Enable barge-in only after echo-trigger rates meet the target.
Evidence	Side-by-side quality/resource benchmark.
Exit condition	A production backend is selected from measured speech recall, false triggers, endpoint latency, CPU, memory, and echo behavior.

12. G7 - Prove STT and Operator Voice Enrollment on Pi 5

Goal: Recognize the words and optionally identify the enrolled operator.

DONE	Record Pi 5 model, CPU, installed RAM, NVMe, and microSD inventory.
TODO	Record free storage capacity and power-supply specification.
TODO	Verify Raspberry Pi OS Lite 64-bit and record the exact version.
TODO	Verify cooling-fan operation, Ethernet, Wi-Fi, and Bluetooth.
TODO	Benchmark sustained CPU temperature and idle/inference power.
TODO	Establish common STT, TTS, and AI-provider adapter interfaces.
TODO	Build fixed STT audio and TTS script evaluation corpora.
TODO	Select and license-review a Pi 5 STT runtime/model.
TODO	Benchmark faster-whisper and/or whisper.cpp on the Pi 5.
TODO	Benchmark STT separately from P4 VAD boundaries.
TODO	Report WER and command accuracy by tone, distance, and noise.
TODO	Install and benchmark Ollama and/or llama.cpp model serving.
TODO	Record model-load time, first-token latency, and sustained token speed.
TODO	Benchmark local TTS and test sherpa-onnx as a replaceable candidate.
TODO	Configure one cloud STT provider and two cloud TTS candidates for measured fallback comparisons.
TODO	Use LM Studio on the Acer workstation for controlled model comparison.
TODO	Select and license-review a pretrained speaker-embedding model.
TODO	Use 12-20 clean enrollment clips from at least two days.
TODO	Keep validation and test clips out of enrollment.
TODO	Average and normalize valid enrollment embeddings.
TODO	Obtain explicit consent before collecting other-speaker trials.
TODO	Export genuine and impostor similarity scores.

TODO	Run <code>evaluate_speaker_scores.py</code> and select a validation threshold.
TODO	Run the untouched final test once.
TODO	Add stronger confirmation for consequential commands.
Evidence	STT report, private enrollment-profile metadata, score report, and selected threshold rationale.
Exit condition	Measured performance is acceptable for the chosen use; voice identity is not the sole authorization for risky actions.

13. G8 - Connect Voice Events to Facial Gestures

Goal: Make the device respond visibly without waiting for the Pi 5.

TODO	Map armed/no-speech to the attentive or idle face.
TODO	Map confirmed speech start to listening.
TODO	Map accepted utterance end to heard.
TODO	Map Pi processing to thinking.
TODO	Drive speaking from actual playback state/amplitude.
TODO	Map timeout, offline, and error conditions explicitly.
TODO	Coalesce obsolete events so display work cannot backlog audio.
TODO	Verify gesture transitions during 100 voice turns.
Evidence	State-event log and recorded demonstration.
Exit condition	Visible state matches the audio/conversation state without causing audio overruns, excessive latency, or stale animation playback.

14. G9 - Concurrent Acceptance and Soak Test

Goal: Prove the complete voice path while the rest of the assistant runs.

TODO	Run microphone, VAD, Wi-Fi, STT exchange, TTS playback, and three LCDs.
TODO	Record both-core utilization and worst-case saturation.
TODO	Record task stack, internal heap, PSRAM, and ring/queue watermarks.
TODO	Count microphone overruns, playback underruns, dropped frames, and resets.
TODO	Run quiet-room false-trigger testing for at least one hour.
TODO	Run a four-hour representative conversation/display soak test.
TODO	Test Pi restart, AP loss, VAD backend failure, and buffer pressure.

TODO	Confirm push-to-talk and offline/degraded behavior still work.
Evidence	Soak-test report and retained metrics.
Exit condition	No watchdog event or unbounded memory growth; zero unexplained audio gaps; failures recover without rebooting the complete assistant.

15. Deferred Backlog

These are explicitly outside the first implementation path:

TODO	wake-word model selection and evaluation;
TODO	low-power LP-I2S hardware VAD experiment;
TODO	full-duplex barge-in before AEC is proven;
TODO	STT model fine-tuning before baseline errors are measured;
TODO	speaker-model training from scratch;
TODO	voice-only authorization for sensitive operations;
TODO	vision-assisted presence or liveness checks;
TODO	compressed audio transport unless PCM measurements justify it.

Optional Jetson Nano evaluation

These items become active only if the Jetson is selected for the vision path:

TODO	Record the exact board/dev-kit revision, CPU/GPU specification, storage, JetPack, and Ubuntu versions.
TODO	Record its power supply, power mode, and cooling arrangement.
TODO	Verify Ethernet/Wi-Fi, CUDA, OpenCV acceleration, and TensorRT.
TODO	Benchmark a representative vision model.
TODO	Measure sustained temperature and power.
TODO	Compare vision latency against the P4 and Pi 5 alternatives before assigning it a permanent architectural role.

16. Development and Merge Gate

Apply this gate to every implementation work package before merge or release:

TODO	Work-package acceptance criteria are satisfied.
TODO	Unit and contract tests pass where present.
TODO	Lint, type, and static checks pass where configured.
TODO	No secrets, private recordings, or biometric data are staged.

TODO	Public interfaces and protocol changes are documented.
TODO	Independent review is complete.
TODO	Review findings are resolved or explicitly accepted.
TODO	Integration-branch tests pass.
TODO	Relevant physical-hardware testing is complete or clearly marked as pending rather than inferred from a successful build.
TODO	This TODO and affected architecture documents reflect any changed assumptions.

17. Definition of Done

The first Project TARS production milestone is complete only when all of the following are true:

TODO	G0 through G9 exit conditions are met or formally waived with evidence.
TODO	PC and P4 results are reported separately.
TODO	VAD, STT, and speaker-verification errors can be attributed separately.
TODO	All model/component versions and licences are recorded.
TODO	Raw voice, embeddings, and private test reports remain outside Git.
TODO	Push-to-talk remains a working fallback.
TODO	Failure and recovery paths have been tested.
TODO	The architecture documents match the implemented task/core allocation.
TODO	Final thresholds come from validation data, not the test split.
TODO	The public repository contains no biometric recordings or embeddings.